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| **Fiscal Year:** | FY 2012 | **Task Last Updated:** | FY 02/22/2013 |
| **PI Name:** | Adams, Gregory R. Ph.D. | | |
| **Project Title:** | Integrated Endurance and Resistance Exercise Countermeasures Using a Gravity Independent Training Device | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | NSBRI | | |
| **Program/Discipline--Element/Subdiscipline:** | NSBRI--Musculoskeletal Alterations Team | | |
| **Joint Agency Name:** | | **TechPort:** | Yes |
| **Human Research Program Elements:** | (1) **HHC**:Human Health Countermeasures | | |
| **Human Research Program Risks:** | (1) **Aerobic**:Risk of Reduced Physical Performance Capabilities Due to Reduced Aerobic Capacity<br>(2) **Muscle**:Risk of Impaired Performance Due to Reduced Muscle Size, Strength and Endurance | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | gradams@uci.edu | **Fax:** | FY 949-824-8540 |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | 949-824-5518 |
| **Organization Name:** | University of California, Irvine | | |
| **PI Address 1:** | Physiology & Biophysics | | |
| **PI Address 2:** | Dept. of Physiology & Biophysics | | |
| **PI Web Page:** | | | |
| **City:** | Irvine | **State:** | CA |
| **Zip Code:** | 92697-0001 | **Congressional District:** | 48 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2007 Crew Health NNJ07ZSA002N |
| **Start Date:** | 06/01/2008 | **End Date:** | 09/30/2012 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NSBRI |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: End date changed to 9/30/2012, per NSBRI information (Ed., 1/17/2013) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Baldwin, Kenneth ( University of California, Irvine )<br>Caiozzo, Vincent ( University of California, Irvine ) | | |
| **Grant/Contract No.:** | NCC 9-58-MA01601 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | Extended space flight as well as existence on Mars will require exercise equipment and training protocols designed to maintain physical fitness and general health. NASA has determined that current flight rated exercise hardware is not appropriate for use on the future Crew Exploration Vehicle (CEV) (JSC--Johnson Space Center--SAT Report 12/06). Studies will investigate protocols designed to maintain both cardiovascular and musculoskeletal fitness using a gravity independent multi-mode exercise device (M-MED), which has been identified by NASA as potential flight hardware. M-MED can provide either high resistance strength- or low resistance endurance-mode exercises.<br>Phase I -ground based integrated strength & cardiovascular (CV) exercise training under normal weight bearing conditions.<br><br>Phases 2&3 - application of this protocol with progressive levels of inactivity.<br><br>Measurements - total body physical work capacity, muscular mass, strength, and sustained muscle endurance (i.e., extravehicular activity, EVA, related issues). CV-related exercise using M-MED "aerobic" mode configuration designed to minimize the time spent in exercise using high power output, short duration interval training. On alternate days, the M-MED will be configured for strength training which has been shown to result in increased muscle strength and size. These studies will validate the efficacy of concurrent endurance and strength training as a high economy approach to flight crew physical fitness, using a scientifically proven exercise modality that has a high probability for use during prolonged space flight missions. This work directly addresses primary requirements in the National Space Biomedical Research Institute (NSBRI) Request for Applications (RFA):<br><br>1. "New, innovative exercise hardware for deployment on CEV and Martian surfaces that provide efficient means for maintenance of aerobic capacity, bone and muscle strength, and endurance with sufficient reserve for contingencies."<br><br>2. "New, innovative exercise protocols that minimize in-flight crew time necessary to maintain aerobic capacity and muscle strength and endurance, and facilitate reserve for contingencies on lunar and Martian missions." |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | To date, this project has demonstrated that a very modest amount of time invested in exercise using the multi-mode exercise device (M-MED) can produce substantial increases in muscle function and cardiovascular fitness. In particular, the fact that M-MED based exercise induces performance gains at many movement speeds suggests that it may be superior to more traditional methods. In total, these findings show that this equipment and these protocols developed specifically for space flight related application may provide a basis for broader use in situations where space and time constraints may limit access to effective exercise. |
| **Task Progress:** | This study is designed to investigate the effectiveness of a new exercise device, multi-mode exercise device or M-MED, for use during long-duration space flights for the maintenance of cardiovascular and musculoskeletal fitness of astronauts. The M-MED is gravity independent and provides both the high force resistance and low force rowing type resistance. To date the results indicate that foot forces in resistance mode exercise on M-MED are similar to those seen using free weight exercises.<br>Electromyography (EMG) studies demonstrate that similar activation levels are seem in key muscles during either M-MED based or free weight resistance exercise. EMG during rowing demonstrate that the thigh, leg, and arm muscles are robustly activated. Endurance mode exercise was also found to stimulate recruitment of the muscles which support the spine. In a recently completed study, 32 subjects (16 male, 16 female) completed 5 weeks of combined resistance and endurance training using the M-MED. Muscle strength increased ~20% while aerobic fitness measured as maximal oxygen consumption (VO2max) increased ~8%. M-MED training increased time to hand grip fatigue by ~70% and leg extension fatigue by ~27%. Thigh muscle cross sectional area increased ~11% as a result of training. Additional deliverables include cross validation of VO2max testing results between the M-MED device and laboratory standard cycle ergometry based testing protocols. As requested by NASA, additional studies have demonstrated that M-MED based exercise can increase the size and strength of the calf and hamstrings muscle groups. These results demonstrate that the M-MED device can be used for both training and physical work capacity testing providing a platform for in-flight assessment.<br><br>In summary, findings to date indicate that the gravity independent M-MED is a viable option for resistance- and endurance-mode exercise during flight and/or planetary exploration thereby addressing two critical risks: 1) Cardiovascular deconditioning; 2) Decreased muscle strength, endurance, and size (atrophy). Endurance mode exercise may have the added benefit of maintaining and, possibly, improving endurance of arm muscles of flight crews as well as aiding in the maintenance of lower back stability and loading during flight and planetary exploration.<br><br>Presentations at workshops and meetings are listed below and also in the Bibliography section:<br><br>Countermeasure Exercise Using a Single, Gravity Independent, Device to Prevent Cardiovascular and Muscular Deconditioning. J. Cotter, T. Owerkowicz, F. Haddad, P. Tesch, V. Caiozzo, G. Adams. American Society for Gravitational and Space Biology Annual Meeting, 2011.<br><br>Resistance and aerobic flywheel training improves muscle strength and aerobic capacity in ambulatory subjects. Owerkowicz T, Cotter JA, Yu AM, Camilon ML, Hoang T, Baker MJ, Pandorf C, Kreitenberg A, Baldwin KM, Tesch PA, Caiozzo VJ, Adams GR. 18th Humans in Space symposium of the International Academy of Astronauts. Houston, TX, 2011.<br><br>Gravity-independent flywheel exercise training improves aerobic capacity and muscle strength in ambulatory subjects. Owerkowicz T, Cotter JA, Tesch PA, Caiozzo VJ, Adams GR. Experimental Biology Washington, DC, 2011<br><br>DNA Methylation is Altered in Human Skeletal Muscle in Response to Exercise Training. Shlomit Radom-Aizik, Fadia Haddad, Tomasz Owerkowicz, Joseph M. Devaney, Eric P. Hoffman, Per A. Tesch, Gregory R. Adams. American College of Sports Medicine Annual Meeting, 2012<br><br>Influence of 10 days of unilateral lower limb suspension and combined exercise training on human vastus lateralis and soleus muscles. J. Cotter, F. Hadad, A. Yu, T. Hoang, M. Baker, P. Tesch, K. Baldwin, V. Caiozzo, G. Adams. Experimental Biology Annual Meeting, 2012.<br><br>The presence and regulation of antisense long non-coding RNA with altered myosin expression in exercising human |

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| | muscle. Clay E. Pandorf, Fadia Haddad, Tomasz Owerkowicz, Kenneth M. Baldwin, Vincent J. Caiozzo, Gregory R. Adams. Experimental Biology, 2012.<br><br>The Effects Of Muscle Unloading With And Without Exercise Countermeasures On Loading Sensitive Cellular And Molecular Biomarkers. G. Adams, J. Cotter, C. Pandorf, F. Haddad, P. Tesch, K. Baldwin,V. Caiozzo. NASA Human Research Program (HRP) Meeting, 2012.<br><br>Counteracting decrements in muscle function and aerobic capacity during unloading utilizing a gravity independent device. Cotter J, Hoang T, Yu A, Tesch P, Caiozzo V, Adams G. Podium presentation at the Annual Meeting for the American College of Sports Medicine, San Francisco, CA, 2012. |
| **Bibliography Type:** | Description: (Last Updated: 11/10/2025) |
| **Abstracts for Journals and Proceedings** | Cotter J, Owerkowicz T, Haddad F, Tesch P, Caiozzo V, Adams G. "Countermeasure Exercise Using a Single, Gravity Independent, Device to Prevent Cardiovascular and Muscular Deconditioning." 27th Annual Meeting of the American Society for Gravitational and Space Biology, San Jose, CA, November 3-6, 2011.<br>27th Annual Meeting of the American Society for Gravitational and Space Biology, San Jose, CA, November 3-6, 2011. Abstract book. , Nov-2011 |
| **Abstracts for Journals and Proceedings** | Owerkowicz T, Cotter JA, Yu AM, Camilon ML, Hoang T, Baker MJ, Pandorf C, Kreitenberg A, Baldwin KM, Tesch PA, Caiozzo VJ, Adams GR. "Resistance and aerobic flywheel training improves muscle strength and aerobic capacity in ambulatory subjects." 18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011.<br>18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011. , Apr-2011 |
| **Abstracts for Journals and Proceedings** | Owerkowicz T, Cotter JA, Tesch PA, Caiozzo VJ, Adams GR. "Gravity-independent flywheel exercise training improves aerobic capacity and muscle strength in ambulatory subjects." Experimental Biology 2011, Washington, DC, April 9-13, 2011.<br>FASEB Journal. 2011 Apr;25(Meeting Abstract Supplement): 668.3. , Apr-2011 |
| **Abstracts for Journals and Proceedings** | Cotter J, Haddad F, Yu A, Hoang T, Baker M, Tesch P, Baldwin K, Caiozzo V, Adams G. "Influence of 10 days of unilateral lower limb suspension and combined exercise training on human vastus lateralis and soleus muscles." Experimental Biology 2012, San Diego, CA, April 21-25, 2012.<br>FASEB Journal. 2012 Apr;26(Meeting Abstract Supplement):1085.4. , Apr-2012 |
| **Abstracts for Journals and Proceedings** | Pandorf CE, Haddad F, Owerkowicz T, Baldwin KM, Caiozzo VJ, Adams GR. "The presence and regulation of antisense long non-coding RNA with altered myosin expression in exercising human muscle." Experimental Biology 2012, San Diego, CA, April 21-25, 2012.<br>FASEB Journal. 2012 Apr;26(Meeting Abstract Supplement):1086.15. , Apr-2012 |
| **Abstracts for Journals and Proceedings** | Radom-Aizik S, Haddad F, Owerkowicz T, Devaney JM, Hoffman EP, Tesch PA, Adams GR. "DNA Methylation Is Altered in Human Skeletal Muscle in Response to Exercise Training." American College of Sports Medicine 59th Annual Meeting, San Francisco, CA, May 29-June 2, 2012.<br>Medicine & Science in Sports & Exercise 2012 May;44(Suppl 2):348. Search: http://journals.lww.com/acsm-msse/toc/2012/05002#-1957448170 , May-2012 |
| **Abstracts for Journals and Proceedings** | Cotter JA, Hoang T, Yu A, Tesch P, Caiozzo VJ, Adams GR. "Counteracting decrements in muscle function and aerobic capacity during unloading utilizing a gravity independent device." American College of Sports Medicine 59th Annual Meeting, San Francisco, CA, May 29-June 2, 2012. Podium presentation.<br>Medicine & Science in Sports & Exercise 2012 May;44(Suppl 2):110. Search: http://journals.lww.com/acsm-msse/toc/2012/05002#-1957448170 , May-2012 |
| **Abstracts for Journals and Proceedings** | Adams G, Cotter J, Pandorf C, Haddad F, Tesch P, Baldwin K, Caiozzo V. "The effects of muscle unloading with and without exercise countermeasures on loading sensitive cellular and molecular biomarkers." 2012 NASA Human Research Program Investigators' Workshop, Houston, TX, February 14-16, 2012.<br>2012 NASA Human Research Program Investigators' Workshop, Houston, TX, February 14-16, 2012. , Feb-2012 |
| **Articles in Peer-reviewed Journals** | Cotter JA, Yu A, Haddad F, Kreitenberg A, Baker MJ, Tesch PA, Baldwin KM, Caiozzo VJ, Adams GR. "Concurrent exercise on a gravity-independent device during simulated microgravity." Med Sci Sports Exerc. 2015 May;47(5):990-1000. http://dx.doi.org/10.1249/MSS.0000000000000483 ; PubMed PMID: 25160844; PubMed Central PMCID: PMC4340828 , May-2015 |
| **Articles in Peer-reviewed Journals** | Owerkowicz T, Cotter JA, Haddad F, Yu AM, Camilon ML, Hoang TN, Jimenez DJ, Kreitenberg A, Tesch PA, Caiozzo VJ, Adams GR. "Exercise responses to gravity-independent flywheel aerobic and resistance training." Aerosp Med Hum Perform. 2016 Feb;87(2):93-101. http://dx.doi.org/10.3357/AMHP.4229.2016 ; PubMed PMID: 26802373; PubMed Central PMCID: PMC4832934 , Feb-2016 |
| **Articles in Peer-reviewed Journals** | Cotter JA, Plaza-Florido A, Adams GR, Haddad F, Scott JM, Everett M, Ploutz-Snyder L, Radom-Aizik S. "Exercise training attenuates the muscle mitochondria genomic response to bedrest." Med Sci Sports Exerc. 2024 Sep 1;56(9):1615-22. https://doi.org/10.1249/MSS.0000000000003457 ; PMID: 38650118; PMCID: PMC11326991 , Sep-2024 |